

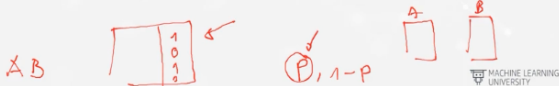
Bayes and Information Theory

Sunday, 18 February 2024 08:56

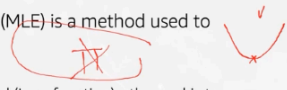
Lecture 4	Lecture 5	Lecture 6
- Derivatives - Gradient Descent - Multivariate Testing - Automatic Differentiation	- Fraud Detection - Bayes Theorem - Naive Bayes - Information Theory	- Presentations - Student presentation of final project - Recommender System

The goal is to find the function that Maximizes the probability of finding a dataset like the ones we are comparing, and use that function to estimate new values that arise

- Maximum Likelihood Estimation (MLE) is a method used to estimate model parameters
 - Propose a model (with its parameters) to capture the underlying relationship in the dataset
 - Compute the Likelihood (joint probability) that this model generates the observed data - the goal is to maximize the Likelihood



- Maximum Likelihood Estimation (MLE) is a method used to estimate model parameters
 - Compute the negative log-Likelihood (Loss function) - the goal is to minimize the Loss function
 - $$-\log(\prod_{i=1}^n p(x_i)) = -\sum_{i=1}^n \log(p(x_i))$$
 - Find the model that minimizes the Loss function (same as maximizing the Likelihood)
 - Compute the derivative of the Loss function
 - Set derivative of the Loss function to zero and solve for the model parameters (that minimize Loss function/maximizes Likelihood)



Multivariate Testing Example

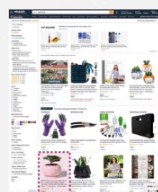
- Process review:
 - Took the derivative :
- $$-\log(f(x_1, x_2, \dots, x_n))' = -k(1 + e^{-\vec{w} \cdot \vec{x}}) + (n - k) \frac{1}{\left(1 - \frac{1}{1 + e^{-\vec{w} \cdot \vec{x}}}\right)}$$
- This is getting messy
 - There's an easier way to finish this process that works for almost all situations

$$L'(\vec{w}) = 0$$

Fraud Detection

- A Multivariable Test finds fraudulent products!
- Fraudulent products tend to have these indicators:
 - Grammatical errors
 - Cheaper than other sellers
 - Different country as buyers
 - Only 1 picture
 - No Reviews

Knowing about fraudulent products, can you find the probability a new product is fraudulent?



Independent Probabilities

- At a roulette table:



- » The last 5 results were:
 - » 14, 5, 30, 18, 12
- » What's the probably the next number is red?
 - » 48.6%
- » The probability of the next number doesn't depend on the previous numbers



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Conditional Probability

- At a single deck Blackjack table:
 - » What's the probability this is an Ace?
 - » There are 4 Aces and 52 cards

$$\frac{4}{52} = \frac{1}{13}$$


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- At a single deck Blackjack table, these hands get dealt
 - » What's the probably the next card is an Ace?
 - » There's already 3 out of 4 Aces showing and 11 cards have been dealt
 - » The probability is **NOT** 1/13 anymore
- When the probability of an event depends on prior events, it's called conditional probability

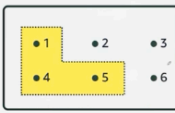


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Probability Diagram

- A **probability space** contains possible outcomes
- A **random variable** holds possible outcomes
- An **event** is a collection of random variables

- » $X \in \{1,2,3,4,5,6\}$
- » $E \in \{1,4,5\}$
- » $P(X \in \{1,4,5\}) = \frac{1}{2}$



$$P_X(E) = \sum_{x \in E} f(x)$$

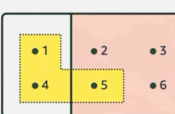
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The way to introduce conditional probability is by restricting the probability space available — for example, what is the probability knowing that only numbers only 1 of the 4 could be it

Probability Diagram

- A **probability space** contains possible outcomes
- A **random variable** holds possible outcomes
- An **event** is a collection of random variables

- » $X \in \{1,2,3,4,5,6\}$
- » $E \in \{1,4,5\}$
- » $C \in \{2,3,5,6\}$



$$P_X(E|C) = P(X \in \{1,4,5\} | \{2,3,5,6\}) = \frac{P(E, C)}{P(C)} = \frac{1}{4}$$

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Bayes Theorem

- Bayes Theorem describes the probability of an event given or conditioned on some other event occurring first

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{\frac{P(A, B)}{P(A)} \cdot P(A)}{P(B)}$$

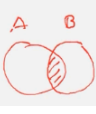
Multiply and divide by P(A)

$$= \frac{\frac{P(B, A)}{P(A)} \cdot P(A)}{P(B)}$$

Intersection is commutative

$$= \frac{P(B|A)P(A)}{P(B)}$$

We know what $P(B, A)/P(A)$ is; namely $P(B|A)$.



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Bayes Theorem Example

- Pick 100,000 product at random

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- » $\frac{1}{1000}$ will have fraud
- » $\frac{9}{10}$ of those with fraud will be flagged
- » $\frac{1}{100}$ will be flagged for fraud when they are not fraudulent

What percentage of flagged products are fraudulent?

Starting number of products	Number of fraudulent products	Number of fraudulent products that are flagged	Number of products that are not fraudulent	Number of products that are not fraudulent but are flagged	Percent of flagged products that are actually fraudulent
100,000	100	90	99,900	999	$\frac{90}{90+999}$

Naïve Bayes

- Can Bayes Theorem to find fraud?
- Looking at data shows fraudulent products tend to have these indicators:
 - » Grammatical errors
 - » Cheaper than other sellers
 - » Different country from buyers
 - » Only one pictures
 - » No reviews

$$P(\text{Fraud} | \text{Grammar}, \text{Cheaper}, \text{Country}, \text{Picture}, \text{Reviews})$$

Naïve Bayes

- Start with a sample of products and aggregate to one variable:

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913loe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
ifreeOw	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2plo134	0	0	0	0	0	0
oiu123i5i	0	1	0	0	0	0
!	!	!	!	!	!	!
ni3899na	0	0	0	0	0	0
nikd12890	1	1	1	0	0	1
ii892kd	0	0	0	0	0	0
hjile389sj	0	0	1	0	0	0
!	!	!	!	!	!	!
8uHb5efv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1
!	!	!	!	!	!	!

	Fraud	Not Fraud
Grammar	3	3
Not Grammar	0	25

Naïve Bayes

- Marginal distribution of counts

	Fraud	Not Fraud	Total
Grammar	3	3	6
Not Grammar	0	25	25
Total	3	28	31

Naïve Bayes

- Joint probability distribution

	Fraud	Not Fraud	Total
Grammar	$\frac{3}{31}=9.7\%$	$\frac{3}{31}=9.7\%$	6
Not Grammar	$\frac{0}{31}=0.0\%$	$\frac{25}{31}=80.6\%$	25
Total	3	28	31

- Marginal probability distribution

	Fraud	Not Fraud	Total
Grammar	3	3	$\frac{6}{31}=19.4\%$
Not Grammar	0	25	$\frac{25}{31}=80.6\%$
Total	$\frac{3}{31}=9.7\%$	$\frac{28}{31}=90.3\%$	31

The probability of it NOT happening is represented with the apostrophe (estilo)

Naïve Bayes

- Marginal probability distribution

	Fraud	Not Fraud	Total
Grammar	3	3	$\frac{6}{31}=19.4\%$
Not Grammar	0	25	$\frac{25}{31}=80.6\%$
Total	$\frac{3}{31}=9.7\%$	$\frac{28}{31}=90.3\%$	31

$$P(F | \dots)$$

$$P(G)$$

$$P(C)$$

$$P(F')$$

Naïve Bayes

- Conditional probability distribution

	Fraud	Not Fraud	Total
Grammar	3	3/28=10.7%	6
Not Grammar	0	25/28=89.3%	25
Total	3	28	31

$$P(G|F') = 10.7\%$$

$$P(G'|F') = 89.3\%$$

Naïve Bayes

- Conditional probability distribution

	Fraud	Not Fraud	Total
Grammar	3/6=0.5	3/6=0.5	6
Not Grammar	0/25=0	25/25=1	25
Total	3	28	31

$$P(F|G) = 0.5$$

$$P(F'|G) = 0.5$$

$$P(F|G') = 0$$

$$P(F'|G') = 1$$

Naïve Bayes

- Getting conditional probability distribution from marginal and joint probability

Conditional probability

	Fraud	Not Fraud	Total
Grammar	3/6=50%	3	6
Not Grammar	0	25	25
Total	3	28	31

Joint probability

	Fraud	Not Fraud	Total
Grammar	3/31	3/31	6/31
Not Grammar	0	25	25
Total	3	28	31

$$P(F|G) = \frac{P(G, F)}{P(G)}$$

$$= \frac{3/31}{6/31} = \frac{3}{6}$$

$$= 50.0\%$$

Marginal probability

We'll do the same calculation, but instead of a single B (grammar) we'll take all variables

Naïve Bayes

- Can Bayes Theorem to find fraud?

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R)}$$

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
nl3993gk	1	0	1	0	0	0
wp9913oe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
fkceOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
olu123i5i	0	1	0	0	0	0
nl389sna	0	0	0	0	0	0
nlkd2890	1	1	1	0	0	1
jl892kd	0	0	0	0	0	0
hyle589sj	0	0	1	0	0	0
8uhb3sefv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R)}$$

- Do not always have enough data for every possible row permutation
- Make a "Naïve" assumption
- Assume columns are independent

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
nl3993gk	1	0	1	0	0	0
wp9913oe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
fkceOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
olu123i5i	0	1	0	0	0	0
nl389sna	0	0	0	0	0	0
nlkd2890	1	1	1	0	0	1
jl892kd	0	0	0	0	0	0
hyle589sj	0	0	1	0	0	0
8uhb3sefv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1

Naïve Bayes

- Independent variables do not have an effect on each other's probabilities

- Ways to mathematically define independence:

$$P(A|B) = P(A)$$

$$\frac{P(A|B)}{P(A)} = 1$$

$$P(A, B) = P(A|B) P(B)$$

$$P(A, B) = P(A) P(B)$$

Naïve Bayes

- Naïve Assumption
 - Assume the variables, once Fraud is held constant, are independent

$$P(G, Ch, Co, P, R|F) = P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)$$

- This assumption doesn't always hold
 - Might be a reason Naïve Bayes underperforms when compared to other methods

Naïve Bayes

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R|F)P(F) + P(G, Ch, Co, P, R|F')P(F')}$$

- The Naïve assumption gives enough to calculate the numerator
- What about the denominator?

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R|F)P(F) + P(G, Ch, Co, P, R|F')P(F')}$$

- The Naïve assumption gives enough to calculate the numerator
- What about the denominator? Use independence assumption too.

$$P(F|G, Ch, Co, P, R) = \frac{P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)P(F)}{P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)P(F) + P(G|F')P(Ch|F')P(Co|F')P(P|F')P(R|F')P(F')}$$

Now it transforms into a pure counting exercise for each variable

- Calculating Bayesian Classifiers

	Fraud	Not Fraud	Totals
Different country	3	7	10
Same country	0	21	21
Totals	3	28	31

$$P(Co|F) = \frac{P(Co, F)}{P(F)} = \frac{3/31}{3/31} = 1 \quad P(Co|F') = \frac{P(Co, F')}{P(F')} = \frac{7/31}{28/31} = 0.25$$

Naïve Bayes

- What the probability of a new product being fraudulent?

Product Id	Grammar	Cheapest	Country	Picture	Review	Fraudulent
7d944ow	1	1	1	0	0	$P(F G=1, Ch=1, Co=1, P=0, R=1)$

$$P(F|G=1, Ch=1, Co=1, P=0, R=0) =$$

$$\frac{P(G=1|F)P(Ch=1|F)P(Co=1|F)P(P=0|F)P(R=0|F)P(F)}{P(G=1|F)P(Ch=1|F)P(Co=1|F)P(P=0|F)P(R=0|F)P(F) + P(G=1|F')P(Ch=1|F')P(Co=1|F')P(P=0|F')P(R=0|F')P(F')}$$

$$\frac{(1)(1)(1)(0.66)(0.33)(0.097)}{(1)(1)(1)(0.66)(0.33)(0.097) + (0.11\%)(0.25)(0.25)(0.93)(1)(0.9)} = 0.792 = 79.2\%$$

Once we detect the probability of a new product, the next step is establishing a threshold above which to alert a human annotator to review — for example, everything above 75% should be reviewed

Information can be defined as the ability to reduce uncertainty

Information Theory

- What is information?

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
h13993jk	1	0	1	0	0	0
wp9913loe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
jfreeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2plo134	0	0	0	0	0	0
olu1235i	0	1	0	0	0	0
lekks90i3	0	0	1	0	0	0
thg39w1i	0	1	0	0	0	0
jfk39e3o	0	0	0	0	0	0
nj229nm3n	1	1	0	1	0	0
o99jnp22	0	0	0	0	0	0
838wkm32	0	1	1	0	0	0
j13ois983	0	0	0	0	0	0

- Information can be thought of as the **resolution of uncertainty**

- Information theory is based on probability theory and statistics

- Information theory often concerns itself with **measures of information of the distributions associated with random variables**

Information Theory

What is information?

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913ioe	0	0	0	0	0	0
9f9925sek	0	1	0	0	0	0
ifkeeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
oiu123isi	0	1	0	0	0	0
ni3899sna	0	0	0	0	0	0
niikd2890	1	1	1	0	0	1
ii892kd	0	0	0	0	0	0
hijle389sj	0	0	1	0	0	0
Buhh3efv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1
...	...	...	...	...	...	...

Information is knowledge about things

What's the probability of a product being fraudulent?

$$P(F) = 3/31$$

What is information?

Product ID	Grammar	Fraudulent
hi3993jsk	1	0
wp9913ioe	0	0
9f9925sek	0	0
ifkeeOwi	0	0
89kj21333	1	1
p2pio134	0	0
oiu123isi	0	0
ni3899sna	0	0
niikd2890	1	1
ii892kd	0	0
hijle389sj	0	0
Buhh3efv	0	0
wd2efv96t	1	1
...	...	...

Information is knowledge about things

What's the probability of a product being fraudulent?

Increase the amount of information by bringing in new variables

We see that the variable grammar provides much more information about whether the object is fraudulent than just having a photo

What is information?

Product ID	Grammar	Fraudulent
hi3993jsk	1	0
wp9913ioe	0	0
9f9925sek	0	0
ifkeeOwi	0	0
89kj21333	1	1
p2pio134	0	0
oiu123isi	0	0
ni3899sna	0	0
niikd2890	1	1
ii892kd	0	0
hijle389sj	0	0
Buhh3efv	0	0
wd2efv96t	1	1
...	...	...

$$3/6 = 50\%$$

$$0/25 = 0\%$$

What is information?

Product ID	Picture	Fraudulent
hi3993jsk	0	0
wp9913ioe	0	0
9f9925sek	0	0
ifkeeOwi	0	0
89kj21333	0	1
p2pio134	0	0
oiu123isi	0	0
ni3899sna	0	0
niikd2890	0	1
ii892kd	0	0
hijle389sj	0	0
Buhh3efv	0	0
wd2efv96t	1	1
...	...	...

$$1/3 = 33\%$$

$$2/28 = 7.1\%$$

Information Theory

Bits

A unit of measurement based on the amount of 1/0 pairs used to encode a message



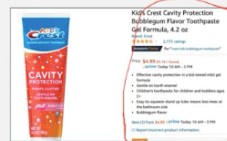
Grammatical Errors

No Grammatical Errors

Information Theory

No grammatical errors is 1 bit of information

Even if the datum has many bytes
Still 1 bit because the original uncertainty could be recoded into 1/0



Grammatical Errors

Have collected at most 1 bit of information

Information Theory

More complex messages

Encode with the minimal possible number of bits



Grammar	Encoding
Misspelling	10
Combined Sentences	01
Misusing word	00
Punctuation	11

At most an average of 2 bits of information

Information Theory

Consider the probability of an outcome when encoding

$$\begin{aligned} \text{Avg(Bit)} &= (12.5\%)^2 \\ &+ (12.5\%)^2 \\ &+ (50.0\%)^2 \\ &+ (25.0\%)^2 \\ &= 2 \end{aligned}$$

Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	10	2
12.5%	Combined Sentences	01	2
50%	Misusing word	00	2
25%	Punctuation	11	2

Average of 2 bits of information at most

Lo que se intenta es reducir al máximo enviar toda la información lo más compacta posible (hoy en día el volumen ya no es un problema). Asignamos a los que tienen mayor peso a los que tienen más frecuencia

Information Theory

- Even though some outcomes have more bits individually, fewer bits on average will be sent

$$\begin{aligned} \text{Avg(Bit)} &= (12.5\%)3 \\ &+ (12.5\%)3 \\ &+ (50.0\%)1 \\ &+ (25.0\%)2 \\ &= 1.75 \end{aligned}$$

Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	000	3
12.5%	Combined Sentences	001	3
50%	Misusing word	1	1
25%	Punctuation	01	2

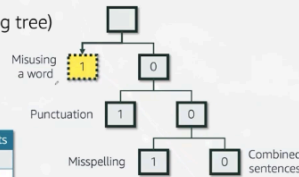
Average of 1.75 bits of information!

What forms is an encoding tree — the most probable should always be placed first going down if they are not met

Information Theory

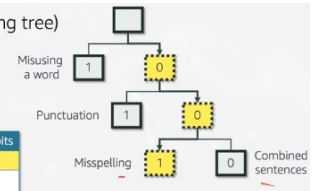
- Encoding Tree (Huffman encoding tree)

Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	001	3
12.5%	Combined Sentences	000	3
50%	Misusing word	1	1
25%	Punctuation	01	2



- Encoding Tree (Huffman encoding tree)

Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	001	3
12.5%	Combined Sentences	000	3
50%	Misusing word	1	1
25%	Punctuation	01	2

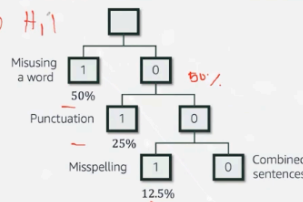


Each split represents a 50% probability

Information Theory

- Encoding Tree

- Each node is a 50/50 split
- A bit corresponds to a single choice
- Bits can be thought of as a unit of randomness

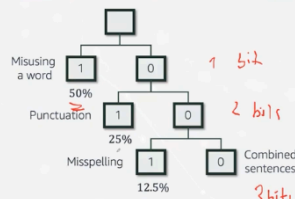


Information Theory

- Encoding Tree

- 1 bit: 50% probability
- 2 bits: 25% probability
- 3 bits: 12.5% probability
- ...

- In general $P = 2^{-N}$
- N is the lengths of bit strings



$$N=2: \frac{1}{4} = 25\%$$

$$N=1: \frac{1}{2} = 0.5$$

Entropy is used to reduce the uncertainty of Decision Trees

Information Theory

- Encoding Tree

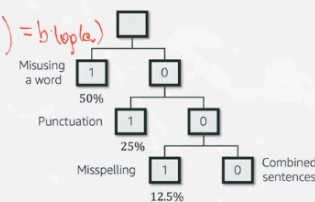
$$P = 2^{-N}$$

- Take the $\log(P)$ to get N

$$-\log_2(P) = N$$

$$E(\text{Bit}) = \sum_{i=1}^k p_i n_i$$

$$\text{Entropy} = -\sum_{i=1}^k p_i \log_2(p_i)$$



- Entropy of a discrete random variable X is a measure of the amount of uncertainty associated with the value of X when only its distribution is known

$$\text{Entropy} = -\sum_{i=1}^k p_i \log_2(p_i)$$

Si toda la información se concentra en una sola posibilidad no hay ninguna entropía, Mientras que cuanto más repartido esté más incertidumbre tendremos.

Information Theory

What is entropy?

» Entropy is a measure of uncertainty, or randomness

A	B	C	D	Entropy
0.25	0.25	0.25	0.25	2
0.5	0.25	0.125	0.125	1.75
0.8	0.1	0.05	0.05	0.31
1.0	0.0	0.0	0.0	0

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Relationship to machine learning

This is all well and good, but how does this relate to machine learning? Well the connections are many, but one of the clearest is in the evaluation of the performance of language models. What these try to do is to learn to generate text in some natural language, like English. The way it does this is that it looks at the text generated such far, and then tries to predict the next character (or potentially several characters or words). It then assigns a probability to each subsequent character. This can be fed into (a slightly modified) definition of entropy so you can understand how many bits of randomness remain after the model tries to make predictions. For many tasks this is the standard unit of measurement, with current top modeling methods getting down to about 0.98 bits per character (BPC).

It doesn't end there, entropy is also key to many techniques in ML such as random forests, and is a cornerstone of understanding how to measure the degree of error a model makes.

Methods to improve our recommendation system from simplest (1) to most complex (7)

⑤ gradient
→ Stochastic grad desc

initiated condition

②

- xavier init
- normal init
- . . .

learning rate

④

- values
- dynamic
- 0.01
- 0.01

③

matrix factorization

$P = A \cdot F$

① latent feature dim

$P = A \cdot F$

$n_u \times k$ $k \times n_b$

data

$S = \frac{1}{r_u}$

0.1 normal

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Dynamic — change the Learning Rate on each pass

Normalize the data: Users in one dimension and rating in the other dimension, scale rating to [0,1] or normal